TABLE 1 **Graph Terminology.**

Type	*Edges*	*Multiple Edges Allowed?*	*Loops Allowed?*
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

Task 1.

1. Draw graph models, stating the type of graph (from Table 1) used, to represent airline routes where every day there are four flights from Boston to Newark, two flights from Newark to Boston, three flights from Newark to Miami, two flights from Miami to Newark, one flight from Newark to Detroit, two flights from Detroit to Newark, three flights from Newark toWashington, two flights from Washington to Newark, and one flight from Washington to Miami, with

a) an edge between vertices representing cities that have

a flight between them (in either direction).

b) an edge between vertices representing cities for each

flight that operates between them (in either direction).

c) an edge between vertices representing cities for each

flight that operates between them (in either direction),

plus a loop for a special sightseeing trip that takes off

and lands in Miami.

d) an edge from a vertex representing a city where a flight

starts to the vertex representing the city where it ends.

e) an edge for each flight from a vertex representing a

city where the flight begins to the vertex representing

the city where the flight ends.

2. What kind of graph (from Table 1) can be used to model a highway system between major cities where

a) there is an edge between the vertices representing

cities if there is an interstate highway between them?

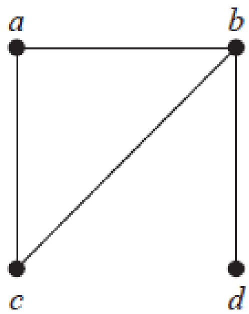
b) there is an edge between the vertices representing

cities for each interstate highway between them?

c) there is an edge between the vertices representing cities for each interstate highway between them, and there is a loop at the vertex representing a city if there is an interstate highway that circles this city?

For **Exercises 3–9**, determine whether the graph shown has directed or undirected edges, whether it has multiple edges, and whether it has one or more loops. Use your answers to determine the type of graph in Table 1 this graph is.

3.



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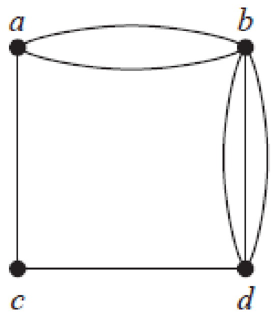

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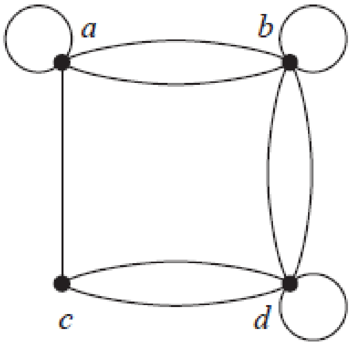

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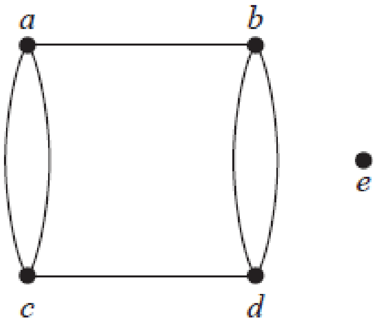

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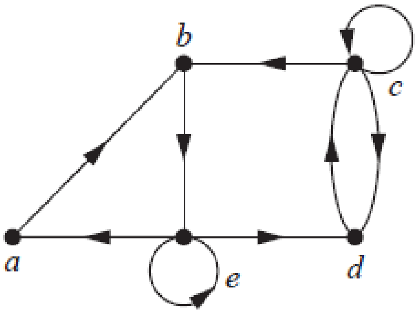

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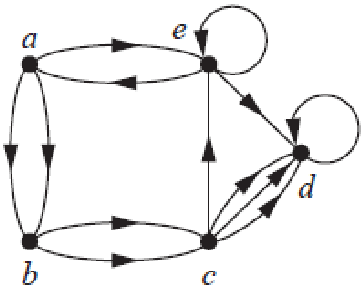

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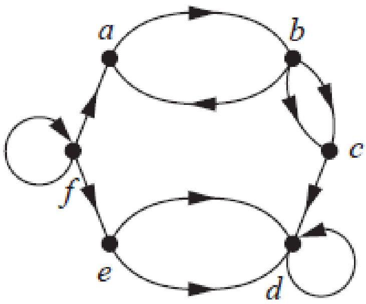

10. For each undirected graph in Exercises 3–9 that is not simple, find a set of edges to remove to make it simple.

11. The **intersection graph** of a collection of sets $A_1, A_2, \ldots, A_n$ is the graph that has a vertex for each of these sets and has an edge connecting the vertices representing two sets if these sets have a nonempty intersection. Construct the intersection graph of these collections of sets.

a) $A_1 = \{0, 2, 4, 6, 8\}$,

$A_2 = \{0, 1, 2, 3, 4\}$,

$A_3 = \{1, 3, 5, 7, 9\}$,

$A_4 = \{5, 6, 7, 8, 9\}$,

$A_5 = \{0, 1, 8, 9\}$

b) $A_1 = \{\ldots, -4, -3, -2, -1, 0\}$,

$A_2 = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$,

$A_3 = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$,

$A_4 = \{\ldots, -5, -3, -1, 1, 3, 5, \ldots\}$,

$A_5 = \{\ldots, -6, -3, 0, 3, 6, \ldots\}$

c) $A_1 = \{x \mid x < 0\}$,

$A_2 = \{x \mid -1 < x < 0\}$,

$A_3 = \{x \mid 0 < x < 1\}$,

$A_4 = \{x \mid -1 < x < 1\}$,

$A_5 = \{x \mid x > -1\}$,

$A_6 = R$

12. Draw the acquaintanceship graph that represents thatTom and Patricia, Tom and Hope, Tom and Sandy, Tom and Amy, Tom and Marika, Jeff and Patricia, Jeff and Mary, Patricia and Hope, Amy and Hope, and Amy and Marika know each other, but none of the other pairs of people listed know each other.

13. Describe a graph model that represents a subway system in a large city. Should edges be directed or undirected? Should multiple edges be allowed? Should loops be allowed?

Task 2.

In Exercises 1–3 find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph. Identify all isolated and pendant vertices.

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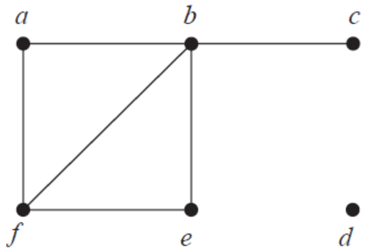

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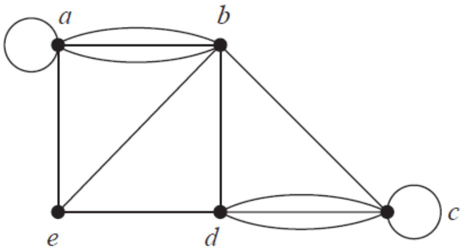

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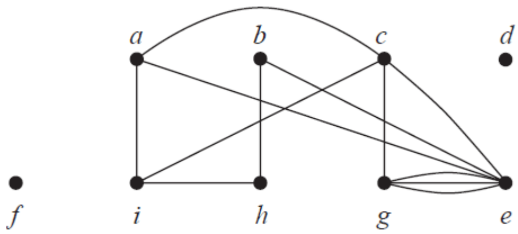


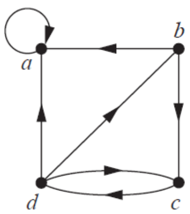
4. Find the sum of the degrees of the vertices of each graph in Exercises 1–3 and verify that it equals twice the number of edges in the graph.

5. Show that the sum, over the set of people at a party, of the number of people a person has shaken hands with, is even. Assume that no one shakes his or her own hand.

6. Show that in a simple graph with at least two vertices there must be two vertices that have the same degree.

In Exercises 7–9 determine the number of vertices and edges and find the in-degree and out-degree of each vertex for the given directed multigraph.

7.



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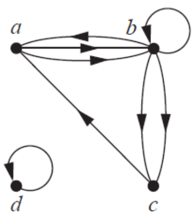

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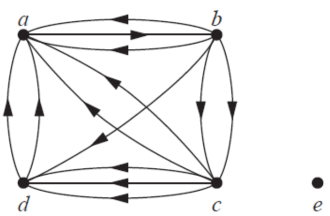

10. For each of the graphs in Exercises 7–9 determine the sum of the in-degrees of the vertices and the sum of the out-degrees of the vertices directly. Show that they are both equal to the number of edges in the graph.

11. Draw these graphs.

a) K_7 **b)** $K_{1,8}$ **c)** $K_{4,4}$

d) C_7 **e)** W_7 **f)** Q_4

12. Suppose that there are five young women and six young men on an island. Each woman is willing to marry some of the men on the island and each man is willing to marry any woman who is willing to marry him. Suppose that Anna is willing to marry Jason, Larry, and Matt; Barbara is willing to marry Kevin and Larry; Carol is willing to marry Jason, Nick, and Oscar; Diane is willing to marry Jason, Larry, Nick, and Oscar; and Elizabeth is willing to marry Jason and Matt.

a) Model the possible marriages on the island using a

bipartite graph.

b) Find a matching of the young women and the young

men on the island such that each young woman is

matched with a young man whom she is willing to

marry.

c) Is the matching you found in part (b) a complete

matching? Is it a maximum matching?

13. Suppose that there are four employees in the computer support group of the School of Engineering of a large university. Each employee will be assigned to support one of four different areas: hardware, software, networking, and wireless. Suppose that Ping is qualified to support hardware, networking, and wireless; Quiggley is qualified to support software and networking; Ruiz is qualified to support networking and wireless, and Sitea is qualified to support hardware and software.

a) Use a bipartite graph to model the four employees and

their qualifications.

b) Use Hall's theorem to determine whether there is an

assignment of employees to support areas so that each

employee is assigned one area to support.

c) If an assignment of employees to support areas so that

each employee is assigned to one support area exists,

find one.